

JAYA SRAVAN KALA

Sunnyvale, CA | 703-389-5214 | jaysravan245@gmail.com | linkedin.com/in/jaysravankala | github.com/jay1197

PROFESSIONAL SUMMARY

Computer Vision & ML Systems Engineer with 3+ years of production experience architecting end-to-end perception platforms, spatial intelligence, and MLOps workflows. Proven success deploying real-time vision pipelines across 400+ physical cameras (6 rollout phases), achieving 97% mAP, reducing duplicate detections by 40%, and cutting manual annotation cycles by 50%. Expert in PyTorch, YOLOv11 (segmentation/OBB), ONNX Runtime edge inference, cross-camera homography projection (Shapely), two-stage GenAI (DOLG visual embeddings + Gemini Flash VLM), and secure distributed event streaming (Kafka, Azure Cosmos DB, BigQuery).

TECHNICAL SKILLS

Languages:	Python, SQL, Bash, JavaScript
Machine Learning & CV:	PyTorch, YOLOv11 (OBB, Segmentation, Detection), Ultralytics, OpenCV, EasyOCR, PaddleOCR, DOLG Visual Embeddings, Gemini Flash VLM, Grounding DINO, Scikit-learn
Spatial & Math:	Shapely (Polygon/IOU Geometry), Homography Projection, NetworkX, RapidFuzz, CheckDigit
MLOps & Edge Inference:	ONNX Runtime, TorchScript, TensorRT, Model Quantization (INT8), MLflow Model Registry, Docker, CI/CD
Distributed & Cloud:	Apache Kafka (Confluent-Kafka, SSL), Azure (Cosmos DB, Blob, Service Bus), Google Cloud BigQuery, AWS S3, REST APIs, Linux/SSH
Data Engineering & Testing:	Pandas, NumPy, PySpark, Delta Lake, PyTest (pytest-cov), SonarQube, Pre-commit Hooks

PROFESSIONAL EXPERIENCE

Sam's Club **Feb 2025 – Present**
Computer Vision Engineer *Sunnyvale, CA*

- Owned the development of production-grade YOLOv11 models (OBB, instance segmentation, and object detection), leading PyTorch training, Optuna hyperparameter tuning, and evaluation to achieve 97% mAP across critical product classes.
- Implemented cross-camera corroboration using floorplan homography projection and Shapely polygon IOU matching, tracking inventory and propagating alerts across physically overlapping camera fields of view.
- Built a high-resolution visual verification pipeline using Grounding DINO and tiled inference on 3840×2160 images, combining EasyOCR with multi-orientation rotation and fuzzy catalog digit correction (RapidFuzz, CheckDigit) to flag 14-digit GTIN mismatches.
- Designed and optimized a production segmentation post-processing pipeline using VSB-guided bin splitting, NMS, and Mask IOU filtering, reducing duplicate and overlapping detections by ~40% across live camera streams.
- Engineered a cost-optimized two-stage change detection pipeline combining 512-dimensional DOLG visual embeddings (cosine similarity) as a fast prefilter with Google Gemini Flash VLM to eliminate false positives from shadows, lighting shifts, and restocking noise.
- Architected an automated image ingestion pipeline integrating enterprise camera streaming APIs with automated quality validation (Laplacian variance blur filtering <100, fisheye lens correction), increasing dataset diversity by 35%, cutting model retraining frequency by 50%, and automating pseudo-labeling via Azure Service Bus to reduce manual annotation time by 50%.
- Integrated an SSL-secured Apache Kafka (Confluent-Kafka) producer to stream real-time inventory alerts to downstream systems, persisting metadata in Azure Cosmos DB and syncing analytics logs to Google Cloud BigQuery; managed model versioning via MLflow Model Registry and exported to ONNX Runtime for low-latency edge inference.
- Hardened production code quality by remediating critical static analysis vulnerabilities (resolving RCE risks and Cosmos DB SQL injection via parameterized @param queries), authoring multi-stage Dockerfiles with automated PyTest coverage suites, and building an internal HTML review tool with CSV export for active learning threshold calibration.

UCode Technologies LLC **Sep 2024 – Jan 2025**
Software Engineer

- Designed and developed ETL pipelines to ingest and transform structured data from SQL databases and AWS S3 into clean, model-ready datasets for predictive analytics.
- Automated data preparation workflows, significantly reducing manual preprocessing effort and enabling faster experimentation for the data science team.
- Developed and evaluated Random Forest classification models, optimizing precision-recall tradeoffs to meet business-specific performance objectives.

Cognizant **Feb 2021 – Jul 2022**
Program Analyst

- Managed and monitored enterprise ETL workflows across MySQL and Snowflake environments, ensuring reliable data availability and SLA compliance.
- Developed SQL transformations and Bash automation scripts using Delta Lake to improve data quality while reducing downstream dashboard processing time.
- Contributed to Snowflake schema optimization and data warehouse enhancements, improving reporting efficiency.

PROJECTS

Multimodal Fake News Detection

- Designed and implemented a multimodal fake news detection system using CLIP and ViLT embeddings to identify inconsistencies between textual and visual content.
- Developed cross-modal similarity analysis using cosine similarity for detecting misleading text-image relationships, effectively identifying fabricated context in news articles.
- Built attention-based interpretability mechanisms to visualize model predictions, improving explainability and trust in the system's output.

Formula One Data Analytics Pipeline

- Architected an end-to-end Azure-based data engineering pipeline using Azure Data Factory, PySpark, and Delta Lake for processing multi-season Formula One datasets.
- Engineered automated data extraction pipelines using REST APIs to ingest live race telemetry and historical driver statistics into Azure Data Lake Storage.
- Optimized PySpark transformations and implemented schema validation and partition optimization, significantly reducing analytical query execution time for downstream dashboards.

EDUCATION

George Mason University <i>Master of Science – Computer Science</i>	Aug 2022 – May 2024 <i>Fairfax, VA</i>
Institute of Aeronautical Engineering <i>Bachelor of Technology – Information Technology</i>	Aug 2017 – May 2021 <i>Hyderabad, India</i>

CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate (DP-203)
- AWS Certified Cloud Practitioner